

Integrated Water System Chapter

Groundwater well, WTP, reserve and operating tanks, nutrient cooling, germination and cooling-pad water

Execution reference for contractor, procurement and multidisciplinary coordination. Physical items shall not be duplicated in other chapters.

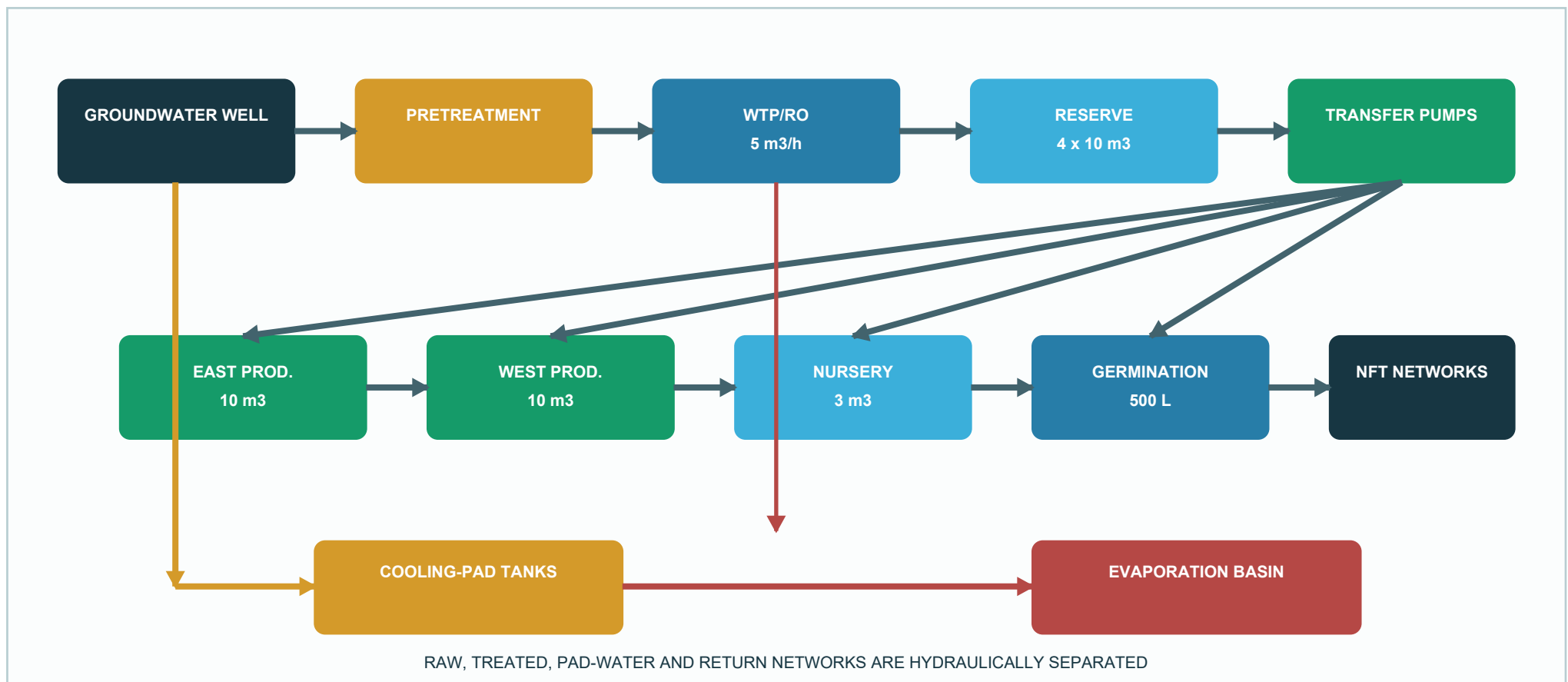
Treatment 5 m3/h treated-water output	Reserve 4 x 10 m3 = 40 m3	Operating 23 m3	Cooling 5 x 7.5 TR = 37.5 TR
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Design basis and system boundaries

Item	Adopted basis	Design note
Project	Production greenhouse 126 x 36 m; service greenhouse 36 x 9 m on the west side	Production is split east/west; service greenhouse is west
Source	Groundwater well	Laboratory analysis is a WTP commissioning hold point
Treatment	5 m ³ /h treated-water output	RO with complete pretreatment and instrumentation
Reserve storage	4 x 10 m ³ = 40 m ³	Unchilled treated water; four tanks in one row
Operating tanks	2 x 10 m ³ + 1 x 3 m ³	East, west and nursery circuits are hydraulically independent
Germination	1 x 500 L	Independent internal tank; no chiller
Cooling-pad tanks	4 x 10 m ³ production + 1 x 1 m ³ service	Separate network; full RO water is not used
Nutrient cooling	5 x 7.5 TR = 37.5 TR	Five identical independent refrigeration circuits

System-boundary rule: This chapter terminates at operating-tank and cooling-pad interfaces. Detailed NFT networks are shown for coordination only and remain quantified in the feed/return chapters; concrete and cabling remain in their own chapters.

Overall architecture and water-quality paths



Well and WTP/RO package

Code	Component	Qty	Capacity/performance	Specification
WTP-01	Complete WTP/RO station	1 set	5 m3/h treated water	Groundwater duty, automatic operation, corrosion-resistant frame
WTP-02	Feed/booster pump	1 in package	Minimum 7 m3/h	Maintains membrane-supplier inlet pressure
WTP-03	Coarse filter / sand separator	1 set	Upstream of pretreatment	Protects filters and pumps from well sand
WTP-04	Multimedia filter	1 set	Sand/multimedia vessel	Reduces turbidity and suspended solids
WTP-05	Iron/manganese filter	1 design allowance	Duty confirmed by analysis	Included for groundwater-source robustness
WTP-06	Antiscalant dosing	1 set	Tank and dosing pump	Upstream of RO membranes
WTP-07	5-micron cartridge filter	1 set	Before high-pressure pump	With differential-pressure indication
WTP-08	High-pressure pump and RO skid	1 skid	Net output 5 m3/h	Includes vessels, membranes and reject connection
WTP-09	Instrumentation and controls	1 set	Pressure, flow and EC/TDS	Raw, permeate and reject monitoring

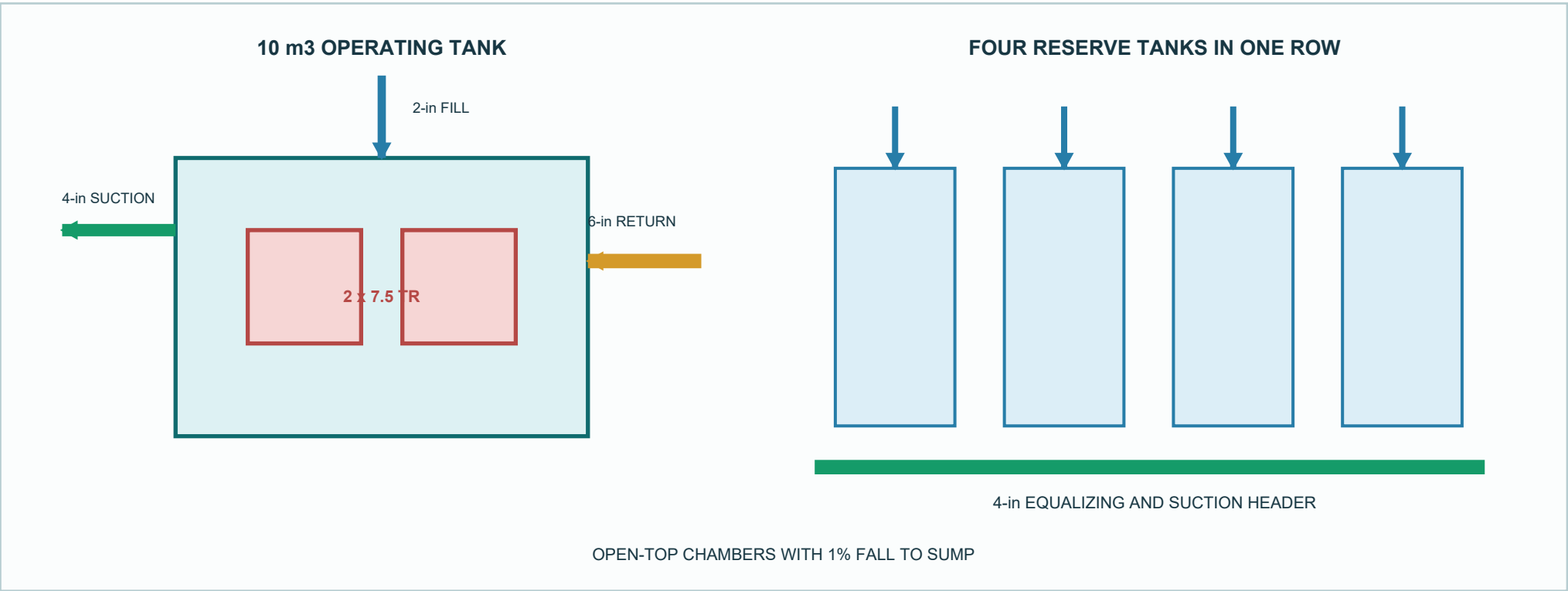
WTP interfaces

2-in raw inlet, 2-in treated outlet, independent reject and wash-drain connections, supplier electrical/control panel, and sample points upstream and downstream.

Acceptance basis

Supplier submits raw-water analysis, membrane projection, recovery, operating pressure, permeate quality, energy/chemical consumption and an 8-hour continuous performance test.

Reserve tanks and chamber



Code	Chamber	Qty	Clear dimensions	Contents	Requirements
RC-RS-01	Reserve-tank chamber	1	12.80 x 3.70 x 3.00 m	Four 10 m3 tanks in one row	Common sump, waterproofing and access ladder

No permanent overflow line is installed. Level control closes the fill valve at high-high. The 4-in low-level equalizing header has a full-bore valve and flexible connector at every tank.

Operating and nursery tanks and chambers

Code	Chamber	Q ty	Clear dimensions	Contents	Requirements
RC-NT-01	Shared divided production-tank chamber	1	7.80 x 4.05 x 3.10 m	Two 10 m3 tanks, pumps and cooling coils	Internal divider, open top, two sumps
RC-NR-01	Nursery-tank chamber	1	3.10 x 3.35 x 2.10 m	3 m3 tank, two pumps and one cooling coil	Open top with sump
RC-PD-01	Production pad-tank chambers	4	4.00 x 4.00 x 3.50 m each	One 10 m3 tank and sector pump per chamber	Independent chamber and sump
RC-PD-02	Service pad-tank chamber	1	2.50 x 2.50 x 2.50 m	1 m3 tank and service pump	Open top with sump

Operating-tank nozzles

Tank	Fill	Suction	Return	Fertigation loop	Bottom drain
East/west production, 10 m3	2 in	4 in	6 in with internal down elbow	1.5-in suction + 1.5-in return	2 in to sump
Nursery, 3 m3	1.5 in	4 in	4 in with internal down elbow	1-in suction + 1-in return	1.5 in

Pump sets and accessories

Code	Service	Qty	Arrangement	Duty point	Power	Connections
P-RSV	Transfer pumps	2	1 duty + 1 standby	12 m ³ /h @ 25 m	2.2 kW / 3 HP	4-in suction; common 2-in discharge
P-NFT-P	Production NFT pumps	3	2 duty + 1 common standby	30 m ³ /h @ 30 m	5.5 kW / 7.5 HP	4-in suction; 3-in discharge
P-NFT-N	Nursery NFT pumps	2	1 duty + 1 standby	8 m ³ /h @ 25 m	1.5 kW / 2 HP	4-in suction; 3-in discharge
P-GER	Germination pump	1	Duty	Light misting for 200 sites	0.37 kW / 0.5 HP	1/2-in internal main
P-PAD-P	Production pad pumps	5	4 duty + 1 stored spare	18-20 m ³ /h @ 10 m	2.2 kW / 3 HP	2.5-in discharge
P-PAD-S	Service pad pumps	2	1 duty + 1 stored spare	5-6 m ³ /h @ 10 m	0.75 kW / 1 HP	1.5-in discharge
P-SUMP	Chamber sump pumps	9	One per adopted chamber/compartment	10 m ³ /h @ 10 m	0.75 kW / 1 HP	2-in discharge to inspection/containment

Standard pump-skid accessories

- Full-bore suction isolation
- Large basket strainer or Y-strainer
- Flexible and dismantling joints
- Suction vacuum/pressure gauge
- Eccentric reducer, flat on top
- Silent discharge check valve
- Full-bore discharge isolation
- 0-6 bar pressure gauge
- Flow meter and sample point
- 1-in bypass back to tank
- 150 mm plinth and anti-vibration pads

NFT feed and gravity-return interfaces

Circuit	Pressure feed	Adopted feed lengths	Gravity return	Adopted return lengths
East production	3-in PN16	28.2633 m main + 4 x 61.133 m	6-in main + two 4-in branches	39.1911 m main + 2 x 61.133 m
West production	3-in PN16	28.2633 + 2 x 43.0575 + 2 x 61.133 m	6-in main + two 4-in branches	39.1911 m main + 43.335 + 61.133 m
Nursery	3-in PN16	1.44 + 55.3059 + 8.037 + 16.0899 m	4-in SN8	4.3 + 37.9 + 16.9 + 16.9 m

Feed requirements

uPVC PN16, compatible solvent-weld system, table inspection boxes, isolation valves and unions, hydrotest at 1.5 times operating pressure.

Return requirements

Heavy-duty SN8 uPVC drainage, sealed sockets, 0.5% minimum and 0.7% preferred production fall, 1% nursery fall, with cleanouts at changes and every 20-25 m.

East production, west production and nursery are fully independent. No return may be cross-connected to another circuit's tank.

Nutrient-solution cooling

Code	Served tank	Capacity	Compressor	Heat exchanger
CH-E-1/2	East production tank, 10 m3	2 x 7.5 TR	Fixed-speed Scroll	Two independent SS316L coils
CH-W-1/2	West production tank, 10 m3	2 x 7.5 TR	Fixed-speed Scroll	Two independent SS316L coils
CH-N-1	Nursery tank, 3 m3	1 x 7.5 TR	Fixed-speed Scroll	One SS316L coil

Control 20°C target, 19-21°C band, 23°C alarm, 25°C critical	Refrigerant R-410A, TXV, filter-drier, sight glass and solenoid	Protection HP/LP switches, suction accumulator, receiver and anti-short-cycle
Piping Exposed copper on galvanized supports; insulated UV-jacketed suction	Installation Independent 150-200 mm concrete base and open-sided canopy	Maintenance Coil lifts from tank top; refrigerant joints remain above liquid

Every circuit requires safe tank level and NFT-pump run proof. Each tank has two PT100 sensors in SS316L pockets: one for control and one for verification/alarm.

Germination tank and unit

Capacity 500 L	Location North service greenhouse	Sites 200	Pump 0.5 HP / 0.37 kW
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Water path A 1-in treated-water branch from the WTP outlet fills the tank from the top. A local pump feeds a 1/2-in main and twelve 16 mm lines to the germination racks.	Operation Timer, 24 V solenoid and manual/auto selector. Four nozzles per level: 48 operating nozzles and 55 procurement quantity including spares.
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Item	Specification	Quantity
Tank	Opaque food-grade PE, sealed lid, cleaning opening and level marks	1
Pump	0.5 HP, suitable for light misting	1
Main line	1/2 in	Rack coordinated
Rack lines	16 mm PE	12
Nozzles	Cleanable fine-mist nozzles	55
Floor drain	To service-greenhouse drain network	At least 1

Cooling-pad tanks and pumps

System	Sectors	Tanks	Pump duty	Pump discharge	Top header	Return	Drain
Production pads	4 sectors	4 x 10 m3	18-20 m3/h @ 10 m	2.5 in	2-in perforated	4 in	1.5 in
Service pads	1 sector	1 x 1 m3	5-6 m3/h @ 10 m	1.5 in	1.25 in	3 in	1 in

Make-up water

From the well after filtration and hardness/scale control, not from full-RO permeate. A 2-in header provides a 15 m3/h interface at 2-3 bar, with 1.5-in production and 1-in service branches.

Collection and discharge

Minimum 200 x 200 mm lower collection gutter at 0.5% fall. Bleed is manual based on EC and discharges through a 2-in line to the lined evaporation basin.

All five sectors are hydraulically independent; no tank or pump is shared between operating sectors.

Controls, protection and drainage

System	Control devices	Protection logic
Reserve level	Low / High / High-High	Transfer lockout at low-low; dry-run protection
Operating-tank filling	One motorized valve per tank	Sequential fill; high-high closes valve and stops transfer
NFT pumps	VFD and run proof	Overload, dry run, phase and pressure/flow protection
Chillers	2 x PT100 per tank	20°C target, 19-21°C band, 23°C alarm, 25°C critical
Cooling permissive	Safe level + NFT pump running	Prevents coil icing and localized cooling
Cooling pads	Fill floats + dry-run protection	Automatic fill; manual EC-based bleed; sectors isolated
Sump drainage	Start float + high alarm	One submersible pump per space with Manual/Auto

Drainage and dewatering

- 600 x 600 x 600 mm sump in each adopted space
- Chamber floor falls 1% toward sump
- 2-in discharge with check valve, isolation and union
- No permanent operating-tank overflow
- Pad bleed and washwater to lined evaporation basin
- RO reject uses an independent approved discharge route
- Service greenhouse has independent 4-in drainage
- 15 central-aisle floor drains at 9 m production bays
- Drainage and pad water never connect to reserve tanks
- All drains have cleanouts and insect protection

Testing and acceptance

Test	Procedure	Acceptance criterion
Network flushing	Flush before instruments and sensors are installed	Clear water with no visible debris
Pressure test	1.5 times operating pressure on pressure networks	No leakage or pressure loss
Gravity return test	Flow and fall test for each circuit	No ponding or reverse flow
Pump test	Measure flow, head and input power	Within $\pm 5\%$ of design duty
WTP performance	Continuous 8-hour operation	5 m ³ /h and permeate quality matching analysis
Tank test	Progressive fill and nozzle/level inspection	No leakage; sensors operate correctly
Chiller test	Staged operation under load	Maintains 19-21°C with logged temperatures
Pad test	Wetting-uniformity test per sector	Uniform wet face without dry zones or overflow
Controls test	Simulate levels and faults	Every interlock and alarm follows cause-and-effect

Mechanical contractor

WTP supplier

Refrigeration supplier

Consultant / Hydrofarms

Equipment, piping and BOQ boundaries

Item	Unit	Qty	Summary specification
WTP/RO station	set	1	5 m3/h including pretreatment and controls
10 m3 reserve tanks	tank	4	Opaque food-grade PE
10 m3 production tanks	tank	2	Opaque insulated food-grade PE
3 m3 nursery tank	tank	1	Opaque insulated food-grade PE
500 L germination tank	tank	1	Inside service greenhouse
10 m3 production pad tanks	tank	4	One per independent chamber
1 m3 service pad tank	tank	1	Independent chamber
7.5 TR refrigeration units	unit	5	R-410A, Scroll and TXV
SS316L immersed coils	coil	5	Top-removable service arrangement
PT100 temperature sensors	sensor	6	Control and verification sensor per operating tank
Pumps across all services	pump	24	2 transfer + 3 production + 2 nursery + 1 germination + 7 pad + 9 sump

Main water-pipe takeoff

Code	Route	Diameter	Centreline	Procurement	Specification
WS-01	Well to WTP	2 in	20 m	22 m	Pressure HDPE/uPVC, isolation and sample point
WS-02	WTP to reserve group	2 in	45 m	50 m	uPVC PN16, flow and EC/TDS measurement
WS-03	Four reserve top-fill branches	1.5 in	12 m	14 m	Valve and flexible connector per tank
WS-04	Reserve equalizing header	4 in	15 m	17 m	Full-bore valve and flexible connector per tank
WS-05	Transfer-pump suction header	4 in	8 m	9 m	Flooded suction, strainer and short straight run
WS-06	Transfer discharge to operating tanks	2 in	60 m	66 m	2-in production and 1.5-in nursery branches
WS-07	WTP branch to germination tank	1 in	45 m	50 m	Treated-water compatible uPVC/PPR
WS-08	Cooling-pad make-up header	2 in	Site coordinated	65 m allowance	1.5-in production and 1-in service branches

No duplication: RC chambers, excavation and waterproofing belong to civil works. Power panels and cables belong to electrical. NFT networks after pump discharge and gravity returns to the tanks belong to feed/return. The fertigation skid and A/B/acid tanks belong to fertigation. This chapter owns the tanks, pumps, chillers and raw/treated/pad-water interfaces.